SIMATS ENGINEERING

ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1596

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3) **Total Marks: 40**
Question Count: 5

Duration : 60 Minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks-I

Experiments

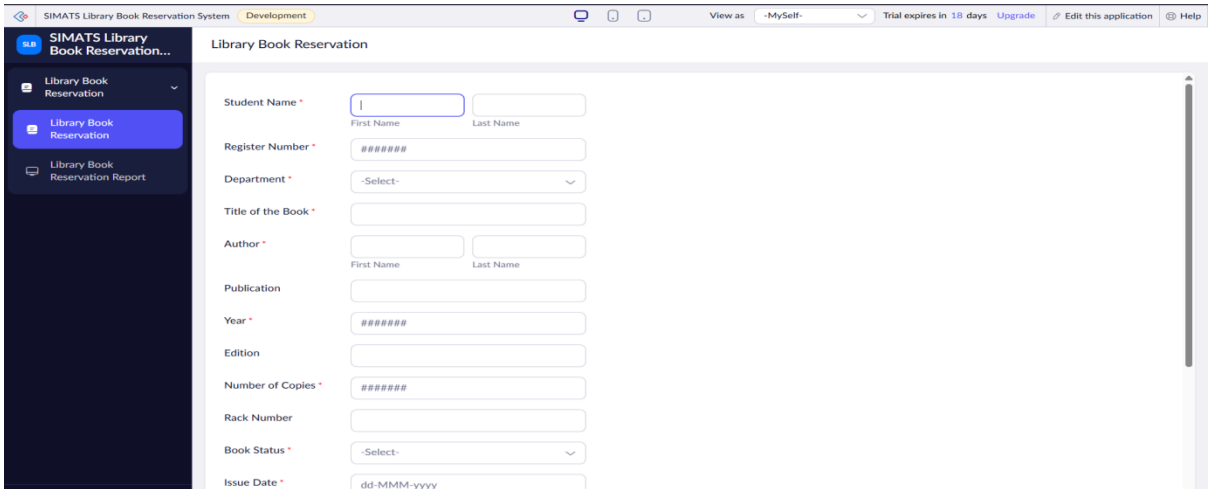
Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

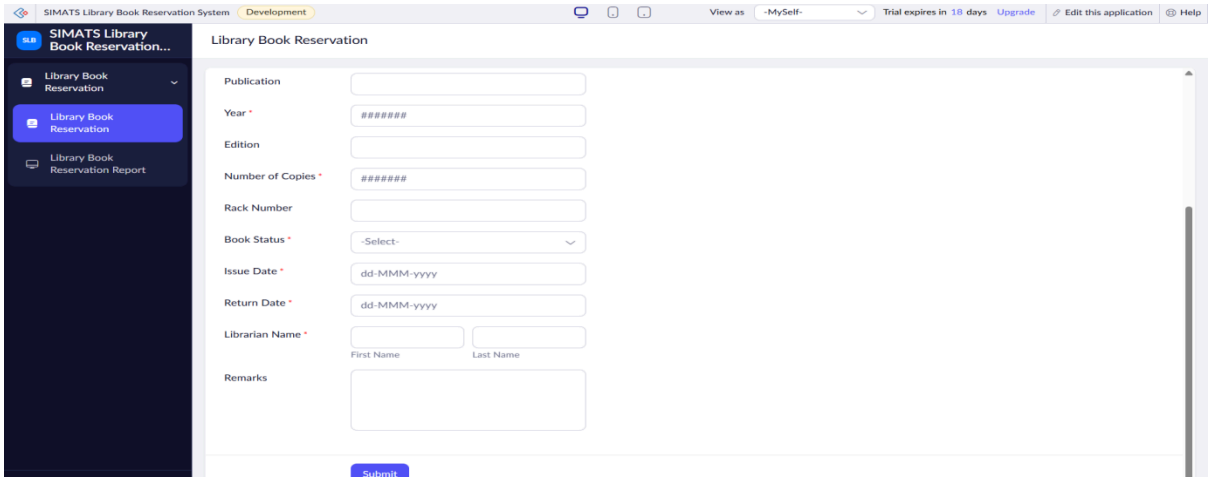
Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

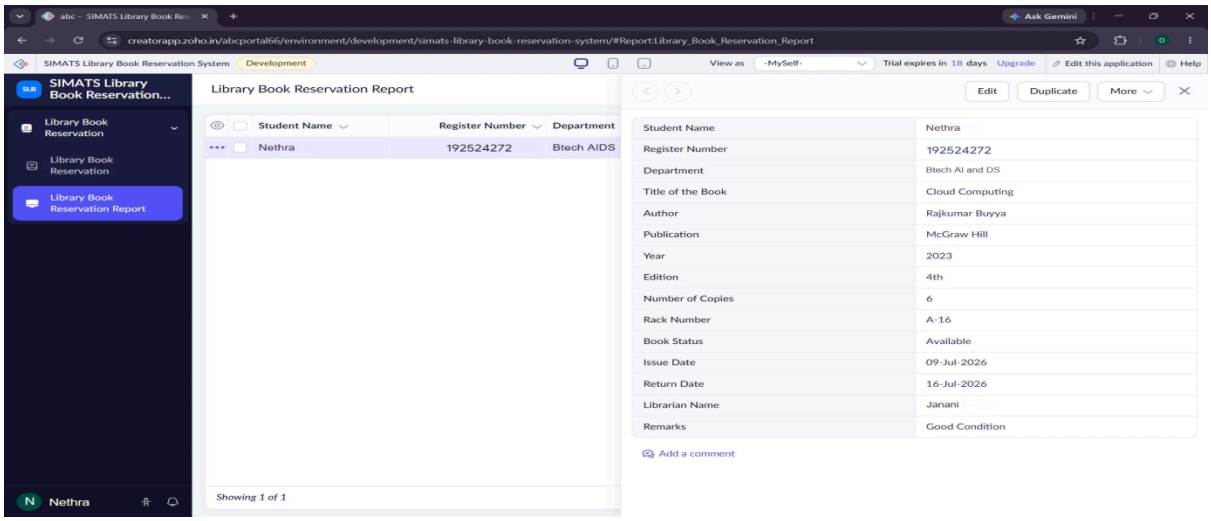
Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

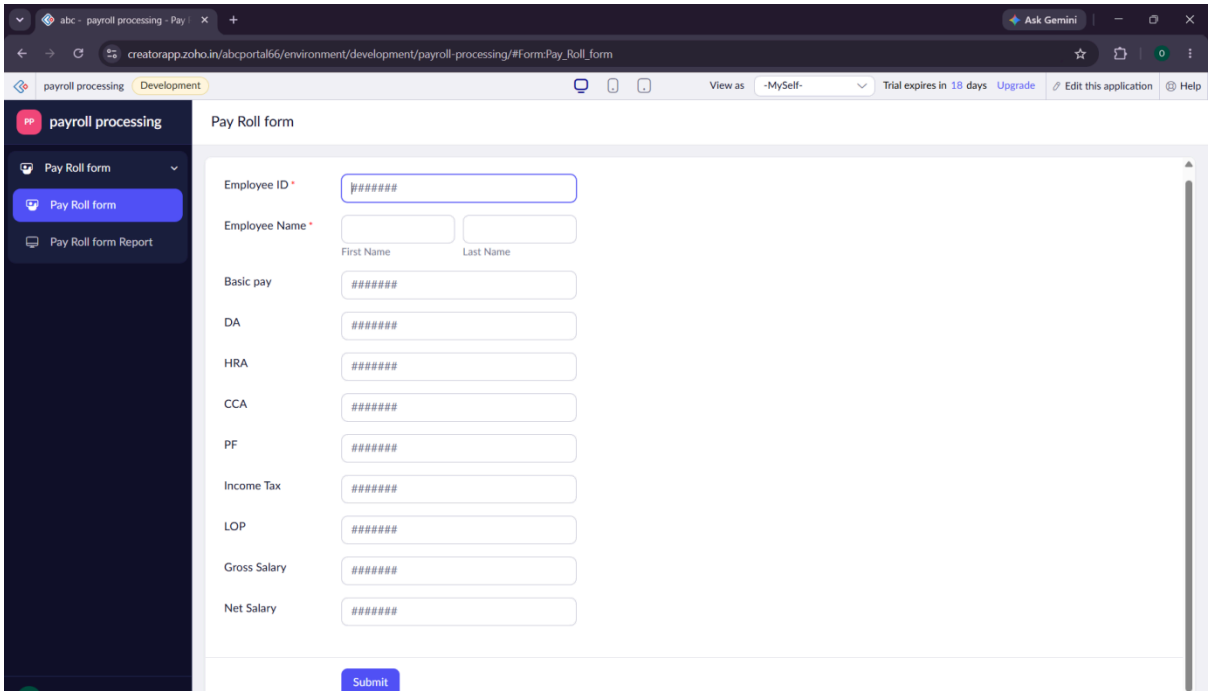
EXPERIMENT -1

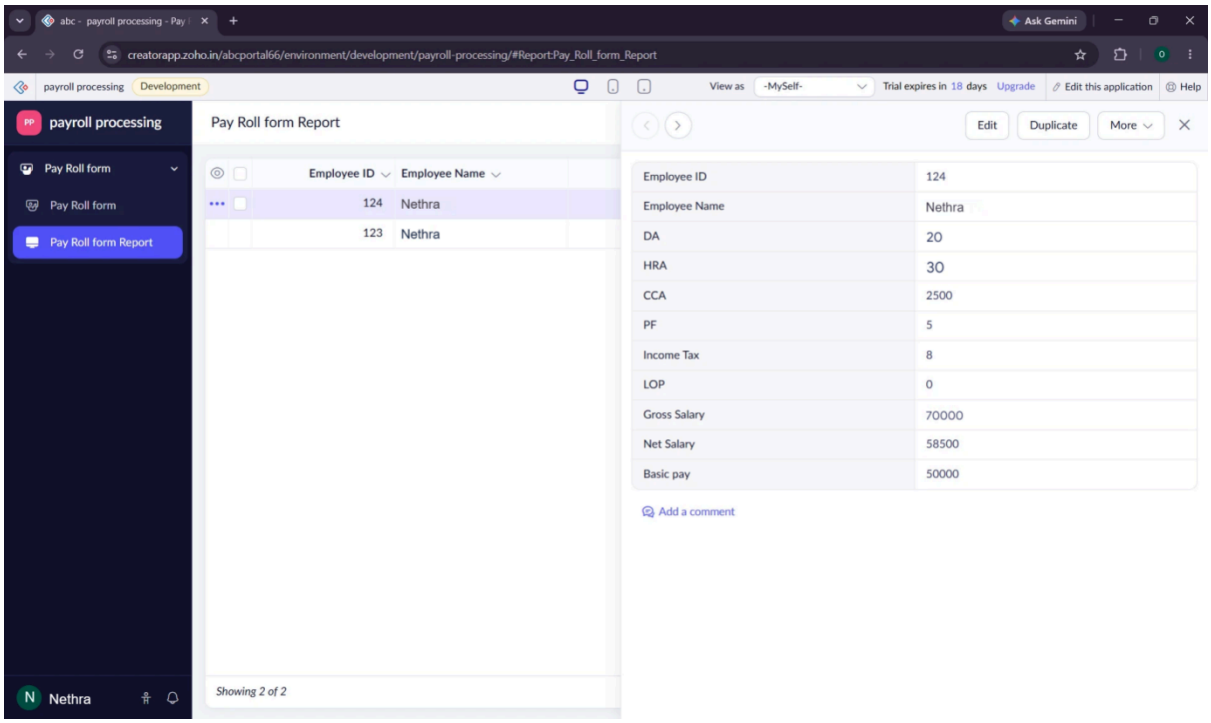






EXPERIMENT-2





EXPERIMENT – 3

Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Aim:

To create a VM with:

- No of processor : 1
- Memory : 2 GB

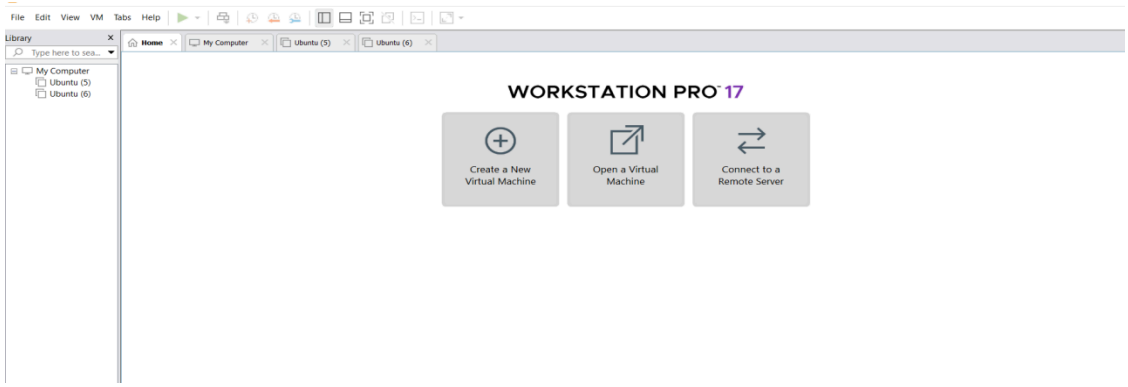
- Hard disk : 15 GB

Steps:

1. Download VMware software and install it as Type-2 Hypervisor.
2. Download Ubuntu or tiny os as ISO image file.
3. In VMware workstation Create new VM
4. Do the configuration based on given conditions.
5. Created a virtual Machine .
6. Launch the VM.

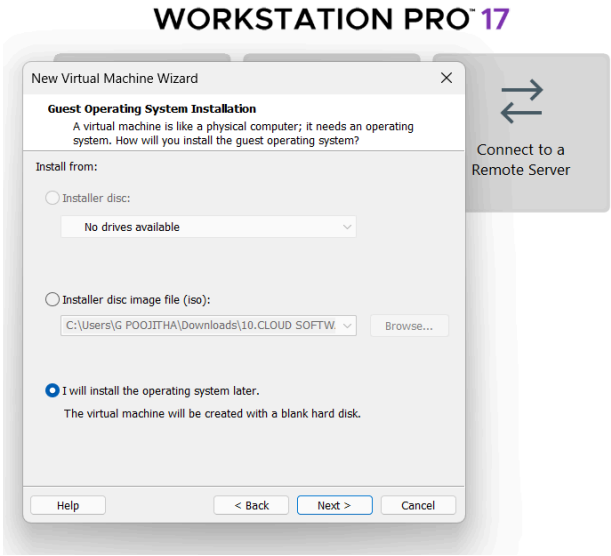
Implementation:

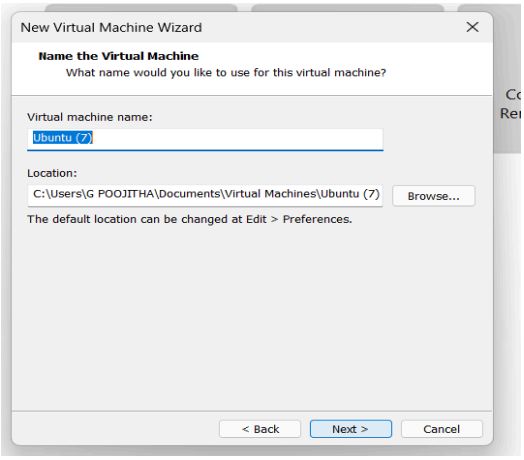
Download and open VMware software.

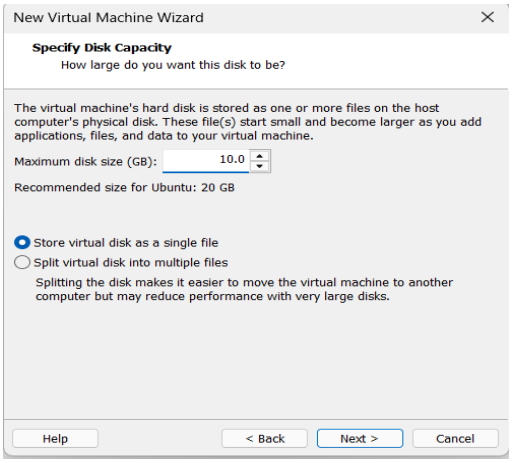


2.Create a new VM

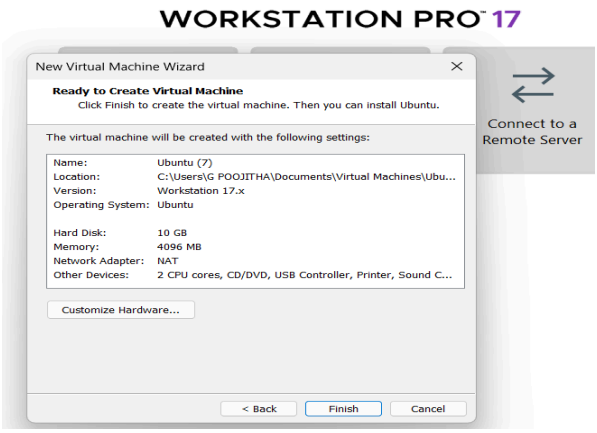






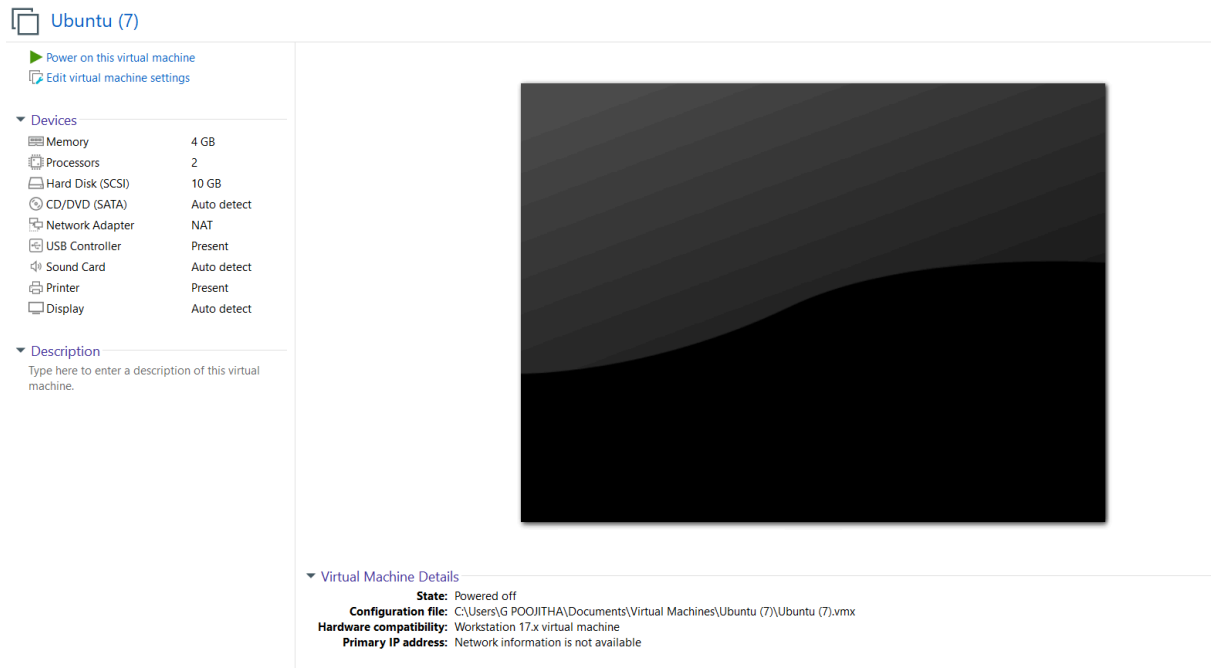


Name it anything□click on next□Adjust the size of disk □ change the option to first and then click on next.

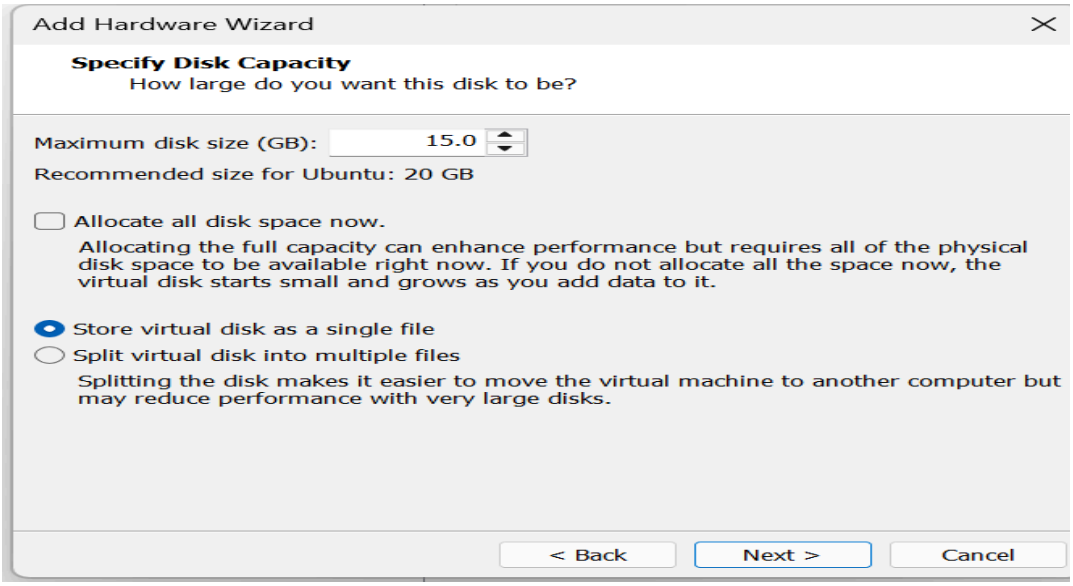


Click on Finish to Launch .

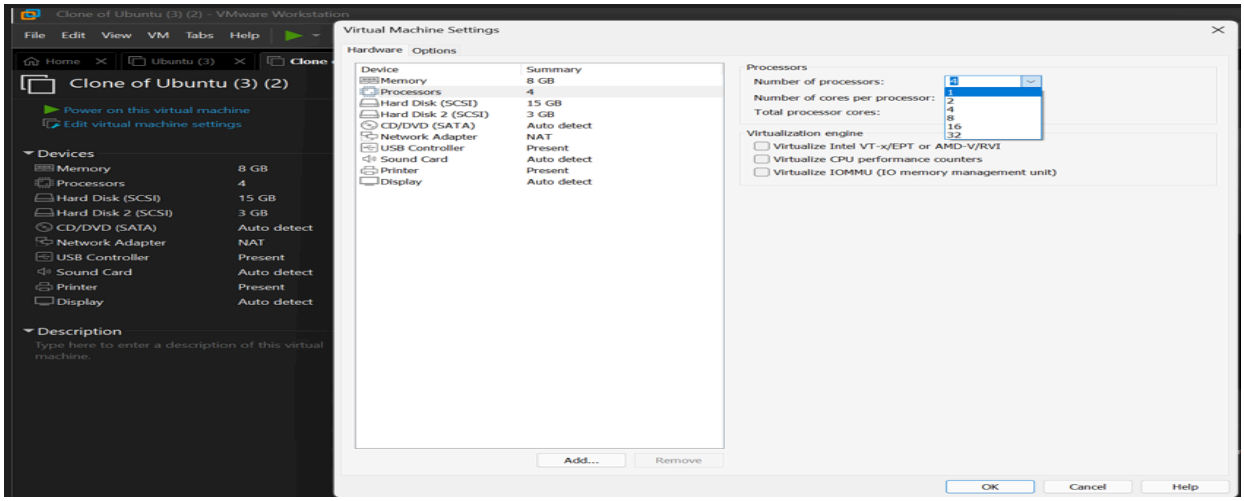
3.VM is created



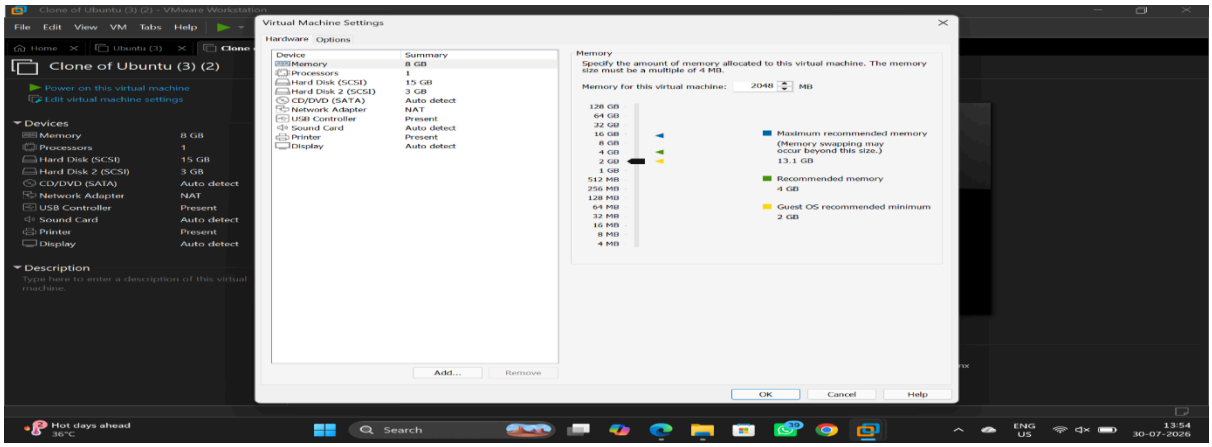
4. we can customize the configuration of hard disk while crearing VM and can add additional hard disk later if needed



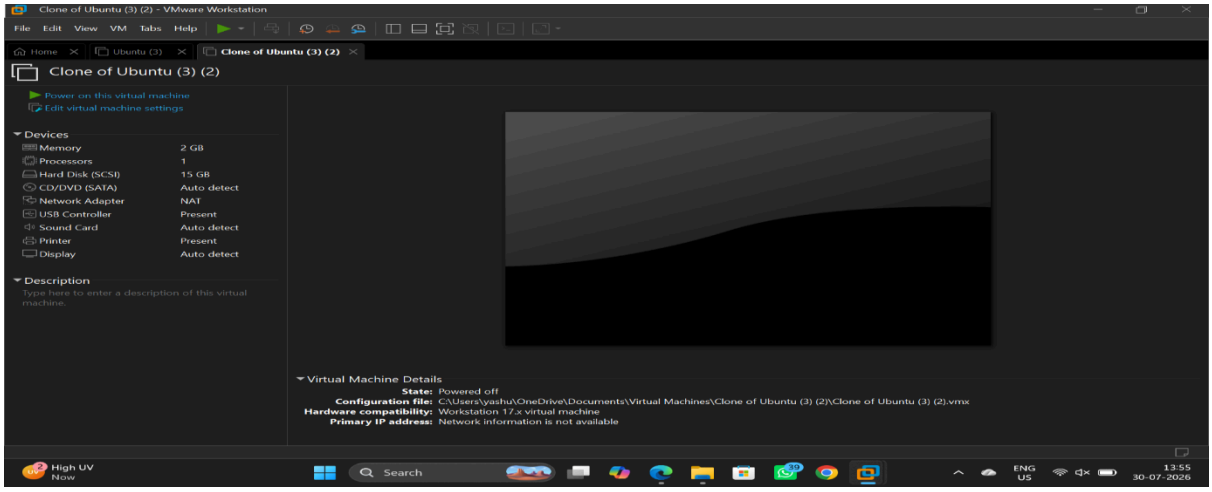
5.Double click on the processor and set it to given configuration



7.Double click on the ram to customize the desired storage



8.VM with given configuration



EXPERIMENT-5

Create a Cloning of a VM and Test it by loading the Previous

Version/Cloned VM

Aim:

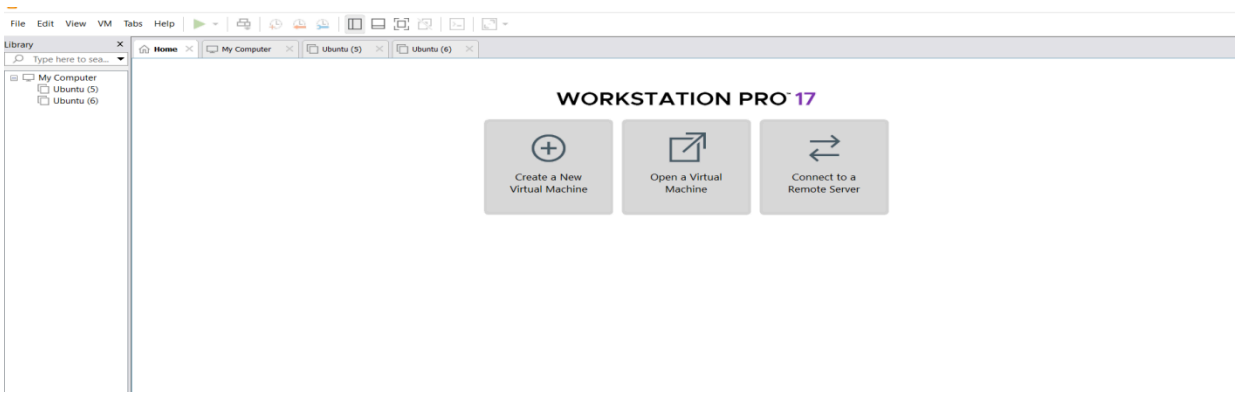
To Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM.

Steps:

7. Download VMware software and install it as Type-2 Hypervisor.
8. Download Ubuntu or tiny os as ISO image file.
9. In VMware workstation Create new VM
10. Do the configuration based on given conditions.
11. Create VM
12. Go to VM Manage Clone.
13. Launch the VM.

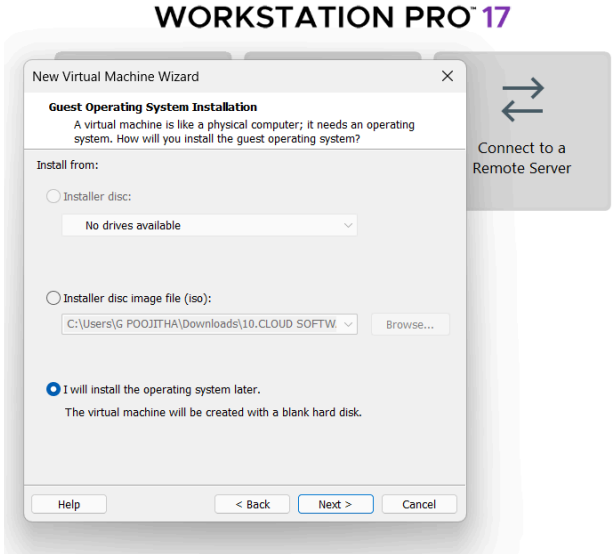
Implementation:

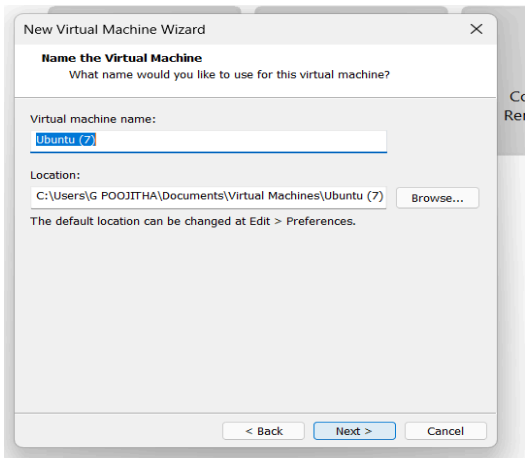
1. **Download and open VMware software.**

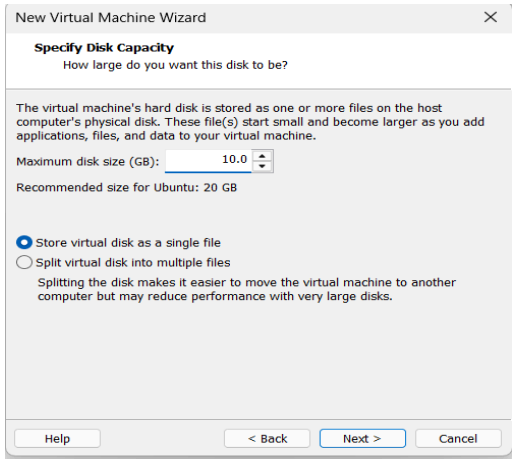


2.Create a new VM

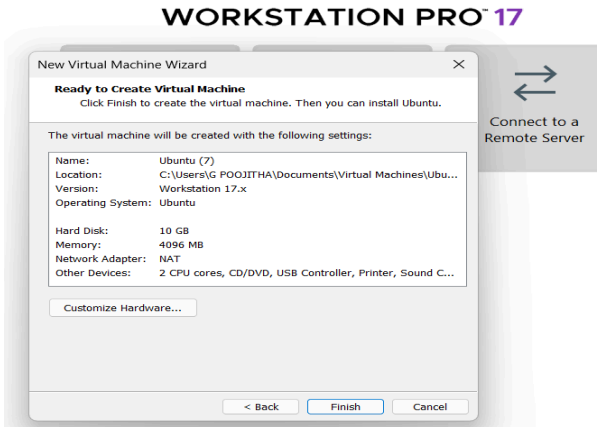






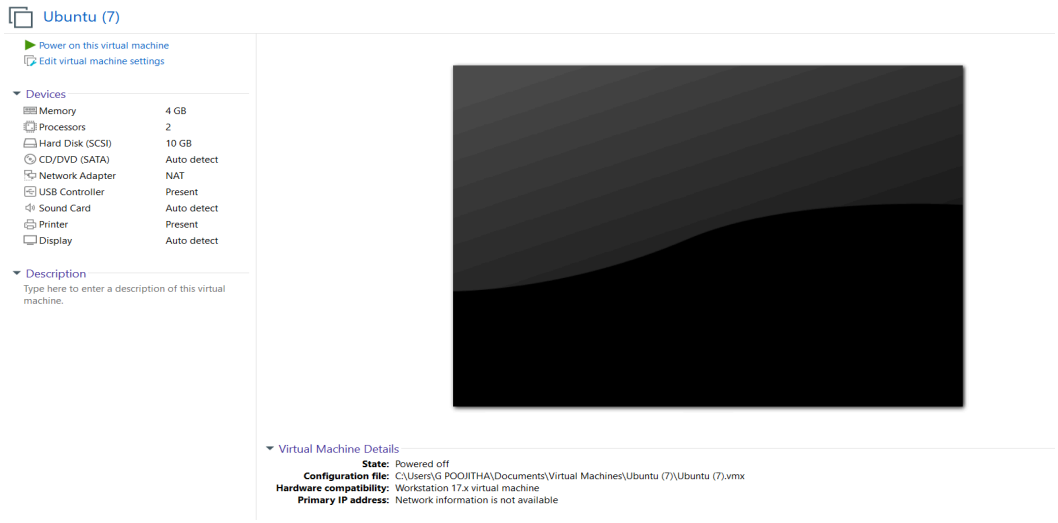


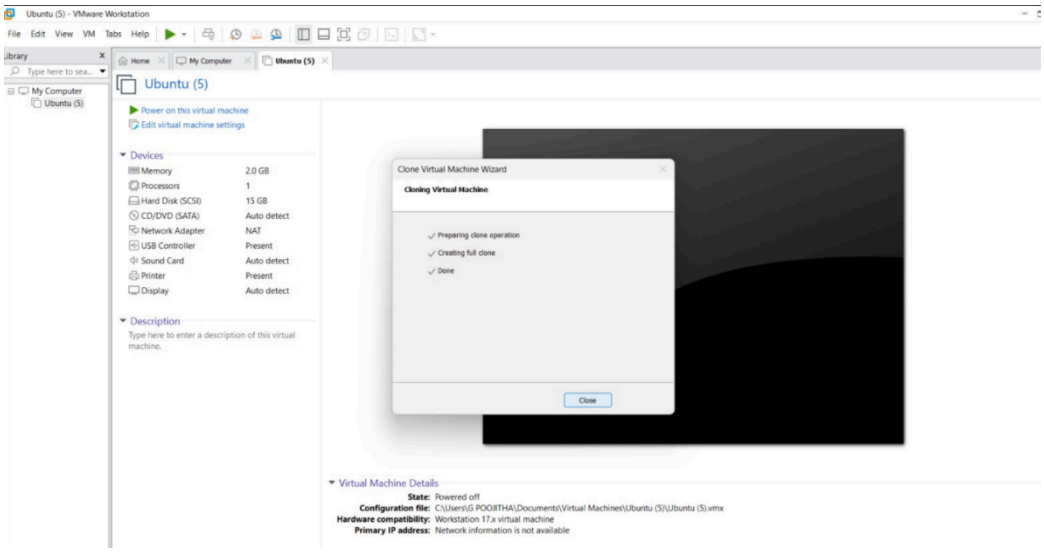
Name it anything□click on next□Adjust the size of disk □ change the option to first and then click on next.



Click on Finish to Launch .

3.VM is created





4. Create Clone

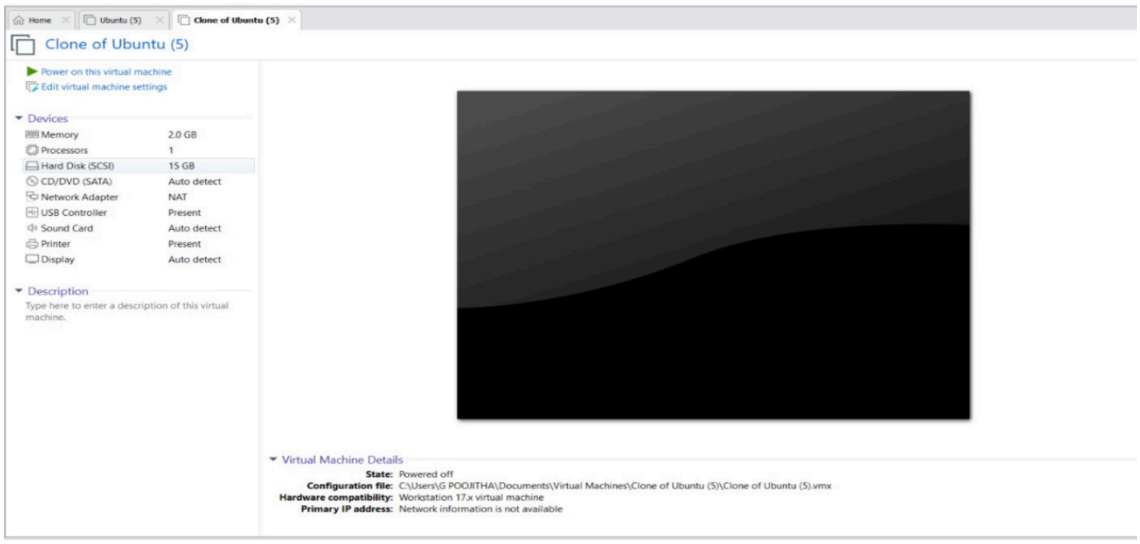


Fig : SSClone has Created and it has same properties as before VM.